

TARGET-DATA2C MVSEL2 HARDEN1 v3 runtime clarification

Release candidate: mdstats 0.20.242a0

Governing chain specification: mlff_target_data2c_mvsel2_forward_lazy_chain_spec.md

Governing workplan: DOC-MVSEL2-HARDEN1-V3 revision 1

Status: candidate runtime contract; acceptance remains qualification/verification-owned

This specification is a narrow normative clarification of the existing MVSEL2/MVSTATE2/REPAIR2 forward-lazy chain. It does not change selector or repair scientific semantics, MVIDX1 scientific identity, target sizes, coverage thresholds, or legacy v1 readability.

Native forward-only campaign boundary

Production MVSEL2 and REPAIR2 runtime construction obtains its MVIDX1 execution view from the authenticated native forward-only record. The v2 selection and repair execution boundary does not require mapping the MVIDX1 witness-to- candidate inverse arrays. Independent qualification and legacy consumers may open the complete MVIDX1 authority outside that measured boundary.

Interrupted MVSEL2 continuation

When a complete target_multi_view_selection_v2 authority is absent, campaign selection searches authorized MVSTATE2 rung checkpoints from highest compatible materializable rank downward. A checkpoint must pass its existing lineage, manifest, array, selected-prefix, and policy validation before use. A corrupt, stale, truncated, or incompatible newest checkpoint is skipped in favor of the next earlier compatible checkpoint; if none is valid, selection rebuilds from rank zero.

Historical TargetMultiViewSelectionEntry and rung evidence preceding the restored rank is reconstructed only by replaying the already-selected candidate prefix through forward rows. Historical candidate choice is not rerun. The authenticated stored FP64 continuation values are retained after structural replay validation so a restart does not create last-bit tie drift. If continuation begins in Phase B, the lazy frontier is rebuilt by one exact deterministic rebase before further candidate choice.

Resumed and uninterrupted execution must produce identical complete entries, rungs, master order, and content digest under the same semantic inputs.

Selector-to-REPAIR2 state reuse

At a materializable repair boundary, REPAIR2 may restore the corresponding compatible MVSTATE2 pure-selector state while the repair order is still equal to the selector order. Missing or invalid state falls back to selected-prefix forward replay only.

After the first accepted repair swap, later pure-selector checkpoints are no longer eligible. The exact repaired mutable state is carried forward through subsequent rungs. A later pure-selector checkpoint must never be reconciled with a diverged repaired order to synthesize repair state.

Execution telemetry distinguishes at least:

- mvstate2 / authenticated checkpoint restoration;
- selected_prefix_forward_replay / exact fallback reconstruction;
- post_divergence_carried_state / repaired-state continuation.

Exact no-copy REPAIR2 proposal evaluation

A rejected REPAIR2 proposal does not clone or mutate the complete TargetMultiViewForwardStateV2. The frozen zero-unique and hard-safe removal admission invariants allow exact pair-specific hypothetical scoring from the current state with bounded reusable scratch:

1. removed-row witnesses are marked with epoch/stamp scratch;
2. hard and coverage replacement frontiers are evaluated from forward rows;
3. correlation balance is corrected for the hypothetical removal;
4. representative gain uses the exact shared-witness multiplicity correction;
5. diversity uses the same hypothetical multiplicity correction;
6. the frozen repair objective and tie hierarchy choose the winner;
7. only the accepted winner mutates authoritative state, exactly once for the deselection and once for the replacement selection, with the replacement score recomputed in the actual post-removal state.

Reusable proposal scratch is execution-only and does not enter scientific identity. Complete accepted swap records and terminal repaired order remain required to equal REPAIR1 on shared oracle fixtures.

Production-scale qualification boundary

Current acceptance requires default-policy REPAIR2 evidence for every materializable fixed-eight rung through 16,384 on the production 36,408-candidate / 165-family graph. Evidence records per-rung wall time, proposal count and shortlist limit, swap count, state restore/replay mode, RSS/peak RSS, no full-state proposal clones, and forward-only/inverse-array status. The campaign integration path remains bounded by StageResourceScope.

The selector + checkpoint/resume + repair chain must retain the governing same-host at-least-10x performance floor versus the frozen MVSEL1 baseline or projection. Missing production input/environment evidence is BLOCKED, not a PASS. GPU qualification remains DEFERRED_NOT_RUN unless actually executed.